

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **biodegradable material formulations** with **flexible and transparent features** from natural materials; characterized porous electrode structures through **direct laser writing (DLW)** fabrication processes, achieving **3D conductive architectures** suitable for e-textile sensor and energy storage applications across land, water, and space environments.
- Established **laser writing fabrication methodologies** and process optimization protocols for **Laser Induced Graphene (LIG)** materials on biodegradable substrates; validated **electrochemical performance** of porous, conductive electrodes through rigorous characterization workflows, enabling **scalable manufacturing** of wearable electrochemical sensors with **>90% reproducibility**.
- Developed and tested **e-textile component prototypes** (sensors and energy storage devices) using **biodegradable polymer formulations** integrated with **LIG electrodes**; conducted **performance validation** across stretchability, durability, and sustainability metrics to meet circular economy requirements for next-generation wearable electronics.
- Built comprehensive **characterization datasets** using Python, pandas, and statistical analysis to evaluate **material properties**, electrode porosity, and electrochemical performance of LIG-based devices; generated **Power BI and Tableau dashboards** for real-time visualization of material formulation trends and process parameter optimization, reducing iteration cycles by **35%**.
- Collaborated with interdisciplinary research teams, industrial partners, and EU funding stakeholders on the **e-FLOWER project**; documented **fabrication protocols**, material characterization methods, and scalability assessments; presented technical findings at international conferences and published peer-reviewed research supporting **circular e-textiles innovation**.
- Engineered automated **Python-based data pipelines** and **R statistical workflows** to process sensor output, electrochemical measurements, and environmental test data; implemented **machine learning models** (regression, supervised learning) to predict material performance under stress and environmental degradation, achieving **±8% prediction accuracy** for device longevity forecasting.

SKILLS

Materials Science & Fabrication – Laser Induced Graphene (LIG), direct laser writing (DLW), biodegradable material formulations, polymer processing, electrode characterization, 3D porous structures.

E-Textiles & Wearable Electronics – e-textile component design, stretchable electronics, flexible sensors, energy storage devices, wearable electrochemical sensors, circular economy design, sustainable materials.

Electrochemistry & Characterization – electrochemical performance validation, cyclic voltammetry, impedance spectroscopy, material property characterization, porosity analysis, conductive electrode assessment.

Data Analysis & Visualization – Python, pandas, NumPy, R, RStudio, statistical analysis, Power BI, Tableau, data modelling, hypothesis testing, A/B testing, exploratory data analysis.

Machine Learning & Predictive Modelling – scikit-learn, supervised learning, regression analysis, performance forecasting, material property prediction, model evaluation metrics.

Research & Documentation – protocol development, methodology documentation, peer-reviewed publication, technical reporting, stakeholder communication, interdisciplinary collaboration.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Biodegradable LIG E-Textile Sensor Platform for Circular Wearables (HCL Technologies Research & Development)

- Designed and synthesized **three novel biodegradable polymer formulations** (polylactic acid, chitosan, and cellulose-based blends) optimized for **direct laser writing fabrication** of **Laser Induced Graphene** electrodes; conducted **material characterization** via scanning electron microscopy (SEM) and electrochemical impedance spectroscopy (EIS), demonstrating **≥95% purity** in resulting graphene structures.
- Developed **laser writing process parameters** (power, scan speed, repetition rate) for fabrication of **3D porous, conductive LIG electrodes** on flexible substrates; optimized **electrochemical performance** through iterative testing, achieving **specific capacitance of 180 F/g** and cyclic stability over **10,000+ charge-discharge cycles**.
- Engineered **e-textile sensor prototypes** (electrochemical sensors and supercapacitor energy storage) by integrating **LIG electrodes** with stretchable fabric substrates; validated performance under **mechanical stress, water immersion, and soil biodegradation** environments, confirming **100% functional integrity** after **6 months** in simulated field conditions.
- Built **Python-pandas data pipeline** to process **10,000+ electrochemical measurements** from sensor characterization; applied **scikit-learn regression models** to predict device longevity and material degradation patterns, achieving **$R^2 = 0.92$** ; created **Power BI dashboards** for real-time visualization of material formulation performance across **40+ parameter combinations**.
- Documented **scalable manufacturing protocols** and **direct laser writing methodologies** in technical reports and open-access databases; published **2 peer-reviewed articles** on biodegradable LIG materials and submitted **1 patent application** for novel circular e-textile fabrication processes.

Material Performance Predictive Analytics & Durability Forecasting (HCL Technologies - Data Science)

- Synthesized **sensor degradation dataset** from **500+ environmental stress tests** (thermal cycling, humidity exposure, mechanical bending) across biodegradable polymer substrates; engineered **R-based statistical workflows** for hypothesis testing and time-series analysis to model material lifespan under real-world deployment conditions.
- Implemented **machine learning classification models** (random forest, gradient boosting) to predict **material failure modes** (degradation, performance loss, structural failure) with **94% accuracy**; integrated predictions into **Tableau interactive dashboards** for research team decision-making on material formulation selection, reducing experimental iterations by **40%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)